NSMO of KSA: Stage 1 Screening & Stage 2 Regional Qualifiers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01 • NSMO of KSA (■■■■■■■■■ ■■■■■ ■■■■■■■ — Mawhiba)

NSMO of KSA: Stage 1 Screening & Stage 2 Regional Qualifiers Practice Exercises

1 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{15} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 5

b. 4

c. 3

d. 15

2 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{24} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 6

b. 4

c. 5

d. 24

3 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{35} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 5

b. 7

c. 6

d. 35

4 [NSMO of KSA — Stage 1 Screening] Evaluate the exact telescoping radical sum $\sum_{k=1}^{48} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

a. 8

b. 7

c. 48

d. 6

5 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{8} \frac{2}{(2k-1)(2k+1)}$.

a. $\frac{16}{15}$

b. $\frac{2}{17}$

c. $\frac{16}{17}$

d. $\frac{32}{17}$

© ClipSAT — clipsat.org | All Rights Reserved

6 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{9} \frac{3}{(2k-1)(2k+1)}$.

a. $\frac{27}{17}$

b. $\frac{27}{19}$

c. $\frac{3}{19}$

d. $\frac{54}{19}$

7 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{10} \frac{4}{(2k-1)(2k+1)}$.

a. $\frac{40}{21}$

b. $\frac{40}{19}$

c. $\frac{4}{21}$

d. $\frac{80}{21}$

8 [NSMO of KSA — Stage 2 Regional] Compute the exact value of the finite series $\sum_{k=1}^{11} \frac{5}{(2k-1)(2k+1)}$.

a. $\frac{55}{21}$

b. $\frac{5}{23}$

c. $\frac{110}{23}$

d. $\frac{55}{23}$

9 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{5} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{30}{31}$

b. $\frac{25}{62}$

c. $\frac{15}{31}$

d. $\frac{1}{31}$

10 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{6} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{42}{43}$

b. $\frac{21}{43}$

c. $\frac{18}{43}$

d. $\frac{1}{43}$

11 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{7} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{28}{57}$

b. $\frac{56}{57}$

c. $\frac{49}{114}$

d. $\frac{1}{57}$

© ClipSAT — clipsat.org | All Rights Reserved

12 [NSMO of KSA — Stage 2 Qualifier] Using the quartic factorization $k^4 + k^2 + 1 = (k^2 - k + 1)(k^2 + k + 1)$, evaluate the exact sum $\sum_{k=1}^{8} \frac{k}{k^4 + k^2 + 1}$.

a. $\frac{72}{73}$

b. $\frac{32}{73}$

c. $\frac{1}{73}$

d. $\frac{36}{73}$

© ClipSAT — clipsat.org | All Rights Reserved